Intermolecular Forces and Their Effect on Physical Properties

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Why does water boil at 100°C while acetone boils at only 56°C? Why does water form beads on a waxy surface while ethanol spreads out? Why does honey flow slowly while acetone evaporates quickly? These differences are not explained by the atoms inside the molecules alone. They are explained by the forces *between* molecules — the **intermolecular forces** of attraction. These forces are much weaker than the chemical bonds that hold atoms together within a molecule, but they have a profound effect on the physical properties of substances: boiling point, viscosity, surface tension, vapor pressure, and molar heat of vaporization. Understanding intermolecular forces allows chemists to predict how substances will behave and to explain why water is uniquely suited to support life.

This reading material uses a diagram to differentiate intermolecular forces from intramolecular forces in liquid water. It uses secondary sources to explain the effect of intermolecular forces on liquid properties such as surface tension, viscosity, vapor pressure, boiling point, and molar heat of vaporization. Finally, it uses a table to compare the intermolecular forces of attraction and properties of water, acetone, and ethanol.

Part 1: Intramolecular vs. Intermolecular Forces

What Are Intramolecular Forces?

Intramolecular forces are the forces that hold atoms together *within* a molecule. These are chemical bonds — covalent, ionic, or metallic. They determine the chemical identity of the substance. Breaking them requires a chemical reaction and a large amount of energy.

Type of Intramolecular Force	Description	Example
Covalent bond	Sharing of electrons between nonmetal atoms	O–H bonds in water
Ionic bond	Transfer of electrons between metal and nonmetal	Na^+Cl^- in sodium chloride
Metallic bond	Sharing of electrons among metal atoms	Fe–Fe bonds in iron

What Are Intermolecular Forces?

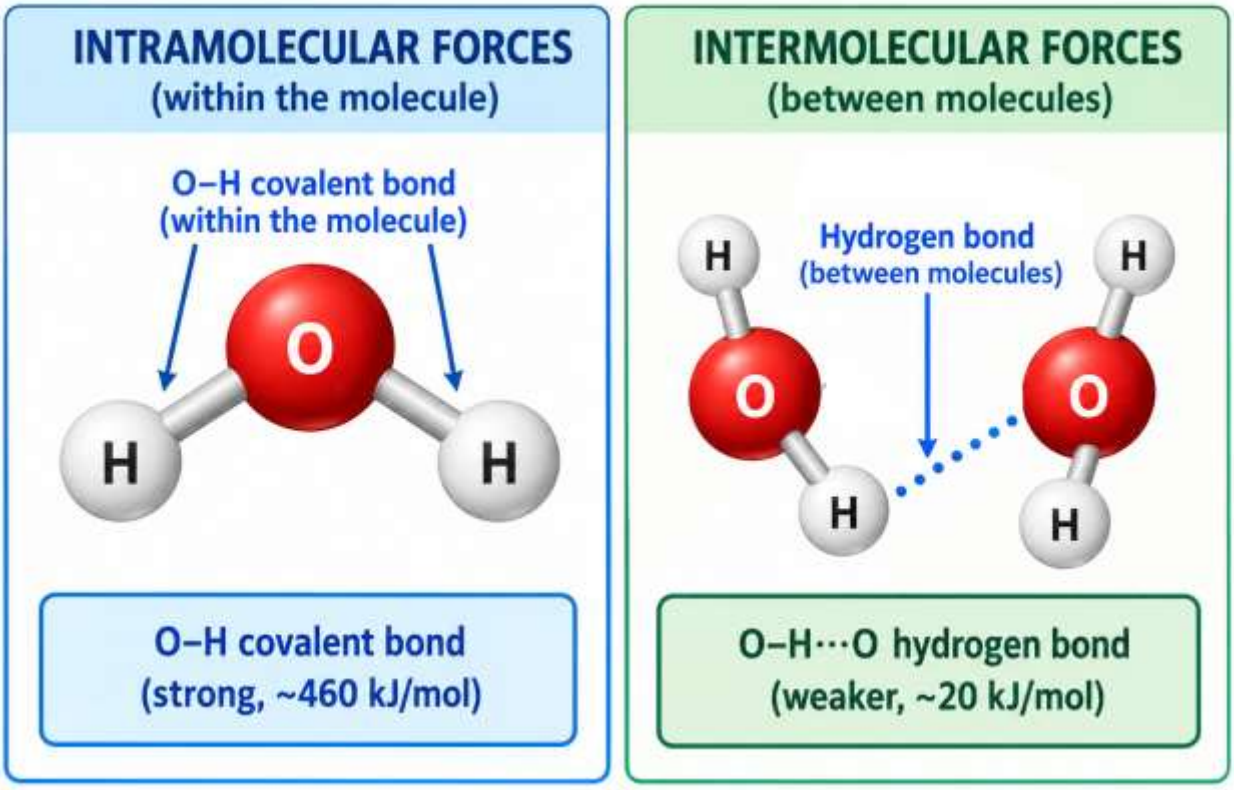
Intermolecular forces (IMFA) are the forces of attraction *between* molecules. They are much weaker than intramolecular forces — typically 1–10% as strong as a covalent bond. They determine the physical properties of a substance: boiling point, melting point, viscosity, surface tension, vapor pressure, and molar heat of vaporization.

Type of Intermolecular Force	Description	Strength (kJ/mol)
London dispersion forces	Temporary dipoles caused by electron movement; present in all molecules	0.05–40
Dipole-dipole forces	Attraction between permanent dipoles of polar molecules	5–25
Hydrogen bonds	Strong dipole-dipole attraction when H is bonded to N, O, or F	10–40

Type of Intermolecular Force	Description	Strength (kJ/mol)
Ion-dipole forces	Attraction between an ion and a polar molecule	40–600

Diagram: Intramolecular vs. Intermolecular Forces in Liquid Water

The diagram below shows liquid water. The **solid lines** represent intramolecular forces (covalent O–H bonds within each water molecule). The **dotted line** represents the intermolecular force (hydrogen bond) between water molecules.



How to interpret the diagram:

Feature	Intramolecular Force	Intermolecular Force
What it holds together	Atoms within a molecule	Separate molecules
Bond type	Covalent bond	Hydrogen bond
Strength	Strong (~460 kJ/mol)	Weak (~20 kJ/mol)
What it determines	Chemical identity	Physical properties
What breaks it	Chemical reaction	Heating, evaporation

Key insight: When water boils, the **intermolecular forces** (hydrogen bonds) break, allowing molecules to escape as gas. The **intramolecular forces** (O–H covalent bonds) do not break. The water molecule remains intact as H_2O in the vapor phase.

Part 2: Effect of Intermolecular Forces on Liquid Properties

1. Surface Tension

Surface tension is the energy required to increase the surface area of a liquid. It is caused by the unbalanced forces on molecules at the surface.

Factor	Explanation
What happens at the surface	Surface molecules are pulled inward and sideways by IMFA but have no molecules above them to pull upward
Result	The surface acts like a stretched elastic film
Effect of IMFA strength	Stronger IMFA → higher surface tension
Example	Water has high surface tension (72 mN/m) due to hydrogen bonds; it forms beads and supports small insects

2. Viscosity

Viscosity is a liquid's resistance to flow. It measures how easily molecules slide past one another.

Factor	Explanation
What causes resistance	Molecules with strong IMFA resist sliding past each other

Factor	Explanation
Effect of IMFA strength	Stronger IMFA → higher viscosity
Effect of molecular size	Larger molecules → higher viscosity (more entanglement)
Example	Honey (strong IMFA and large molecules) flows slowly; acetone (weak IMFA and small molecules) flows quickly

3. Vapor Pressure

Vapor pressure is the pressure exerted by a vapor in equilibrium with its liquid at a given temperature.

Factor	Explanation
What causes vapor pressure	Molecules at the surface escape into the gas phase
Effect of IMFA strength	Stronger IMFA → lower vapor pressure (molecules are held more tightly)
Effect of temperature	Higher temperature → higher vapor pressure
Example	Acetone (weak IMFA) has high vapor pressure and evaporates quickly; water (strong IMFA) has lower vapor pressure

4. Boiling Point

Boiling point is the temperature at which a liquid's vapor pressure equals atmospheric pressure.

Factor	Explanation
What boiling requires	Enough energy to overcome IMFA and allow molecules to escape
Effect of IMFA strength	Stronger IMFA → higher boiling point
Effect of molecular size	Larger molecules → higher boiling point (more electrons = stronger dispersion forces)
Example	Water (hydrogen bonds) boils at 100°C; acetone (dipole-dipole) boils at 56°C

5. Molar Heat of Vaporization

Molar heat of vaporization (ΔHvap) is the energy required to vaporize one mole of a liquid at its boiling point.

Factor	Explanation
What it measures	Energy needed to overcome IMFA completely
Effect of IMFA strength	Stronger IMFA → higher molar heat of vaporization

Example	Water has a high ΔHvap (40.65 kJ/mol) because hydrogen bonds must be broken; acetone has a lower ΔHvap (31.3 kJ/mol) because dipole-dipole forces are weaker

Summary Table: Effect of IMFA Strength on Properties

Property	Stronger IMFA	Weaker IMFA
Surface tension	Higher	Lower
Viscosity	Higher	Lower
Vapor pressure	Lower	Higher
Boiling point	Higher	Lower
Molar heat of vaporization	Higher	Lower

Why This Matters

Application	How IMFA Explains It
Water's high boiling point	Allows liquid water to exist on Earth
Water's high heat of vaporization	Allows sweating to cool the body
Water's high surface tension	Allows capillary action in plants

Application	How IMFA Explains It
Acetone's quick evaporation	Makes it useful as a solvent and nail polish remover
Ethanol's intermediate properties	Makes it useful as a solvent and fuel additive

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Part 3: Comparison of Water, Acetone, and Ethanol

Structures and Intermolecular Forces

Substance	Formula	Structure	Polarity	Types of IMFA Present	Strongest IMFA
Water	H_2O	Bent	Polar	Hydrogen bonds, dipole-dipole, London dispersion	Hydrogen bonds
Acetone	C_3H_6O (CH_3COCH_3)	Trigonal planar at C=O	Polar	Dipole-dipole, London dispersion	Dipole-dipole
Ethanol	C_2H_5OH (CH_3CH_2OH)	Tetrahe-dral at C	Polar	Hydrogen bonds, dipole-dipole, London dispersion	Hydrogen bonds

Note: Acetone cannot form hydrogen bonds with itself because it has no O–H bond. It can only accept hydrogen bonds from other molecules. This is why its IMFA are weaker than water's or ethanol's.

Comparison Table: Properties of Water, Acetone, and Ethanol

Property	Water (H_2O)	Acetone (C_3H_6O)	Ethanol (C_2H_5OH)
Molar mass (g/mol)	18.02	58.08	46.07
Strongest IMFA	Hydrogen bonds	Dipole-dipole	Hydrogen bonds
Boiling point (°C)	100.0	56.1	78.4
Viscosity (mPa·s at 25°C)	0.89	0.31	1.07
Surface tension (mN/m at 25°C)	72.0	23.7	22.4
Vapor pressure (kPa at 25°C)	3.17	30.8	7.9
Molar heat of vaporization (kJ/mol)	40.65	31.3	38.56

Analysis of the Data

Observation	Explanation
Water has the highest boiling point	Water has the strongest IMFA (hydrogen bonds in a 3D network)

Observation	Explanation
Acetone has the lowest boiling point	Acetone has the weakest IMFA (dipole-dipole only)
Water has the highest surface tension	Water's hydrogen bonding network creates strong cohesive forces at the surface
Ethanol and acetone have similar surface tension	Both have weaker IMFA at the surface compared to water
Water has the highest molar heat of vaporization	More energy is needed to break water's extensive hydrogen bonding network
Acetone has the highest vapor pressure	Weak IMFA means molecules escape easily into the gas phase
Ethanol has the highest viscosity	Larger molecules and hydrogen bonding create resistance to flow

Why Water Is Unique

Property	Value	Why It Matters
Boiling point	100°C	Allows liquid water to exist on Earth
Heat of vaporization	40.65 kJ/mol	Allows sweating to cool the body
Surface tension	72 mN/m	Allows capillary action in plants

Property	Value	Why It Matters
Specific heat capacity	4.18 J/g°C	Stabilizes Earth's climate

Key Terms

Term	Definition
Intramolecular force	Force that holds atoms together within a molecule (covalent, ionic, metallic bond)
Intermolecular force (IMFA)	Force of attraction between molecules
London dispersion force	Weak IMFA caused by temporary dipoles; present in all molecules
Dipole-dipole force	IMFA between permanent dipoles of polar molecules
Hydrogen bond	Strong dipole-dipole attraction when H is bonded to N, O, or F
Surface tension	Energy required to increase the surface area of a liquid
Viscosity	A liquid's resistance to flow
Vapor pressure	Pressure exerted by a vapor in equilibrium with its liquid
Boiling point	Temperature at which vapor pressure equals atmospheric pressure
Molar heat of vaporization	Energy required to vaporize one mole of a liquid at its boiling point

Review Questions

1. Differentiate intramolecular forces from intermolecular forces. Provide two examples of each.

2. Using the diagram of liquid water, explain the difference between the O–H covalent bond and the hydrogen bond between water molecules.

3. Explain how intermolecular forces affect each of the following properties:

- (a) Surface tension
- (b) Viscosity
- (c) Vapor pressure
- (d) Boiling point
- (e) Molar heat of vaporization

4. Complete the following table comparing water, acetone, and ethanol:

Property	Water	Acetone	Ethanol
Strongest IMFA			
Boiling point (°C)			
Viscosity (mPa·s)			
Surface tension (mN/m)			
Vapor pressure (kPa)			
Molar heat of vaporization (kJ/mol)			

5. Explain why acetone has a lower boiling point than water. Use the concept of intermolecular forces in your answer.

6. Explain why ethanol has a higher viscosity than acetone even though both have similar surface tension.

7. Why is water's high molar heat of vaporization important for living organisms?

8. Pag-aralan ang sitwasyon sa ibaba at sagutin ang mga tanong.

A student spills water and acetone on a laboratory table. Both are at room temperature (25°C). The student notices that the acetone evaporates much faster than the water.

- (a) Which substance has the higher vapor pressure? Explain.
- (b) Which substance has the stronger intermolecular forces? Explain.
- (c) Predict which substance has the higher boiling point. Explain your reasoning.

9. Using secondary sources, write a short explanation (3–5 sentences) of how intermolecular forces affect the boiling point, viscosity, and surface tension of liquids.

10. Application question: A chemist needs to choose a solvent for a reaction that requires a liquid with a high boiling point and low vapor pressure. Between water, acetone, and ethanol, which would be the best choice? Explain your answer using the data from the comparison table.

Answers

1. Intramolecular forces hold atoms together within a molecule (e.g., O–H covalent bond in water, Na–Cl ionic bond in salt). Intermolecular forces attract separate molecules to each other (e.g., hydrogen bonds between water molecules, dipole-dipole forces between acetone molecules).

2. The O–H covalent bond is an intramolecular force that holds the hydrogen atom to the oxygen atom within a single water molecule. It is strong (~460 kJ/mol) and requires a chemical reaction to break. The hydrogen bond is an intermolecular force that attracts the hydrogen of one water molecule to the oxygen of another. It is weaker (~20 kJ/mol); and breaks when water boils.

3.
(a) Stronger IMFA → higher surface tension.
(b) Stronger IMFA → higher viscosity.

(c) Stronger IMFA → lower vapor pressure.
(d) Stronger IMFA → higher boiling point.
(e) Stronger IMFA → higher molar heat of vaporization.

4.

Property	Water	Acetone	Ethanol
Strongest IMFA	Hydrogen bonds	Dipole-dipole	Hydrogen bonds
Boiling point (°C)	100.0	56.1	78.4
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5. Acetone has only dipole-dipole forces, while water has hydrogen bonds. Hydrogen bonds are stronger than dipole-dipole forces, so more energy is required to break them and allow water molecules to escape as gas. This is why water boils at 100°C while acetone boils at 56°C.

6. Ethanol has a higher viscosity because its molecules are larger and can form hydrogen bonds. Acetone has smaller molecules and only dipole-dipole forces. Viscosity depends on both molecular size and IMFA strength, so ethanol's larger size and hydrogen bonding create more resistance to flow.

7. Water's high molar heat of vaporization means that a large amount of energy is required to convert liquid water to vapor. This is important for living organisms because sweating removes a large amount of heat from the body, helping to regulate temperature.

8.
(a) Acetone has the higher vapor pressure because it has weaker intermolecular forces, allowing molecules to escape into the gas phase more easily.
(b) Water has the stronger intermolecular forces because it has hydrogen bonds, while acetone has only dipole-dipole forces.
(c) Water has the higher boiling point because stronger intermolecular forces require more energy to overcome.

9. Intermolecular forces determine the physical properties of liquids. Stronger IMFA result in higher boiling points, higher viscosity, and higher surface tension. This is because more energy is required to overcome the attractive forces between molecules. For example, water has strong hydrogen bonds, giving it a high boiling point, high surface tension, and high viscosity compared to acetone, which has weaker dipole-dipole forces.

10. Water would be the best choice because it has the highest boiling point (100°C) and the lowest vapor pressure (3.17 kPa) among the three liquids. Its strong hydrogen bonds make it resistant to evaporation and require high energy to boil. Acetone has a low boiling point and high vapor pressure, making it unsuitable. Ethanol has intermediate properties but still lower boiling point and higher vapor pressure than water.